

Master Thesis
BlueRov2 Formation Control

Week 4

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Plan For Fourth Week

- Formation control with single integrator agents no constraints
  - Undirected graph
  - Directed graph
- Single integrator formation control with constraints
  - Only connectivity maintenance
  - Then also colision avoidance
- Double integrator agents

- BlueRov2 model no input constraints
  -
- [Research challenge] integrate input constraints
  -
- [Research challenge] controller robust or less sensitive to uncertainties (e.g. inverse optimal controller in the sense of LgV like control laws rather than feedback linearization)

Clarification

$$\nabla W(\tilde{z}) = \left[\left(\frac{1}{\Delta_k^2 - |z_k|^2}\right) - \left(\frac{1}{|z_k|^2 - \delta_k^2}\right)\right] z_k$$

Code

Code

$$max\left(\Delta_k^2 - |z_k|^2,\, 0.01\right)$$

$$max\left(|z_k|^2 - \delta_k^2,\, 0.01\right)$$

Example:

	$\Delta_k = 50\ ;\ \delta_k = 5\ ;\ \lvert z_k\rvert = 100$	$\Delta_k = 50\ ;\ \delta_k = 5\ ;\ \lvert z_k\rvert = 2$
$\Delta_k^2 - \lvert z_k\rvert^2$	**-7500**	**2496**
$max\left(\Delta_k^2 - \lvert z_k\rvert^2,\, 0.01\right)$	**0.01**	**2496**
$\lvert z_k\rvert^2 - \delta_k^2$	**9975**	**-16**
$max\left(\lvert z_k\rvert^2 - \delta_k^2,\, 0.01\right)$	**9975**	**0.01**

Achievements

- Study
  - Graph Theoretic Methods in Multiagent Networks
  - Esteban’s thesis
  - Khalil's book on Nonlinear Control

- Implementation
  - Improvements
    - Parameters set on terminal
  - Implemented double-integrator agents

Improvements - Parameters set on terminal

```
zen@zen-HP-Pavilion-Gaming-Laptop-15-ec1xxx:~/KTH-Thesis/BlueROV2_Formation_Control/Software$ python3 formation_simulation.py --help
usage: formation_simulation.py [-h] [--N N] [--dim DIM] [--graph GRAPH] [--constraint CONSTRAINT] [--dynamics DYNAMICS] [--dt DT] [--steps STEPS] [--c1 C1] [--c2 C2] [--gamma GAMMA] [--v_max V_MAX]
                               [--a_max A_MAX] [--d_max D_MAX] [--d_min D_MIN] [--R R] [--l L]

Formation Control Simulation Parameters

options:
  -h, --help            show this help message and exit
  --N N                 Number of agents
  --dim DIM             Dimension (2D or 3D)
  --graph GRAPH         Graph type (0: Undirected, 1: Directed)
  --constraint CONSTRAINT
                        Constraints (0: None, 1: Conn, 2: Both)
  --dynamics DYNAMICS   Dynamics (0: Single, 1: Double)
  --dt DT               Size of simulation steps
  --steps STEPS         Number of simulation steps
  --c1 C1               Barrier function gain
  --c2 C2               Control gain
  --gamma GAMMA         Velocity offset gain
  --v_max V_MAX         Maximum velocity
  --a_max A_MAX         Maximum acceleration
  --d_max D_MAX         Connectivity Maintenance threshold
  --d_min D_MIN         Collision Avoidance threshold
  --R R                 Formation radius
  --l L                 Square side length
```

```
zen@zen-HP-Pavilion-Gaming-Laptop-15-ec1xxx:~/KTH-Thesis/BlueROV2_Formation_Control/Software$ python3 formation_simulation.py
Condition Met: Spanning tree exists and system is stable.

==================================================
      BLUE ROV2 FORMATION CONTROL SIMULATION
==================================================
--- CONFIGURATION ---
Agents (N):         10
Dimension:          3D
Dynamics:           Double Integrator
Graph Type:         Directed
Constraints:        Both

--- PHYSICS & CONTROL ---
Time Step (dt):     0.01s (1000 steps)
Gains:              c1=1.0, c2=1.0, γ=2.0
Limits:             v_max=100m/s, a_max=200m/s²

--- GEOMETRY ---
Thresholds:         d_max=50m, d_min=5m
Formation:          Radius=20.0m, Spawn Area=100m²
==================================================
```

Double-integrator agents

Single-integrator agents **VS** **Double-integrator agents**

$x_i \in \mathbb{R}^n$ is the state of each agent

$$\dot{x}_i = u_i, \quad i \leq N,$$

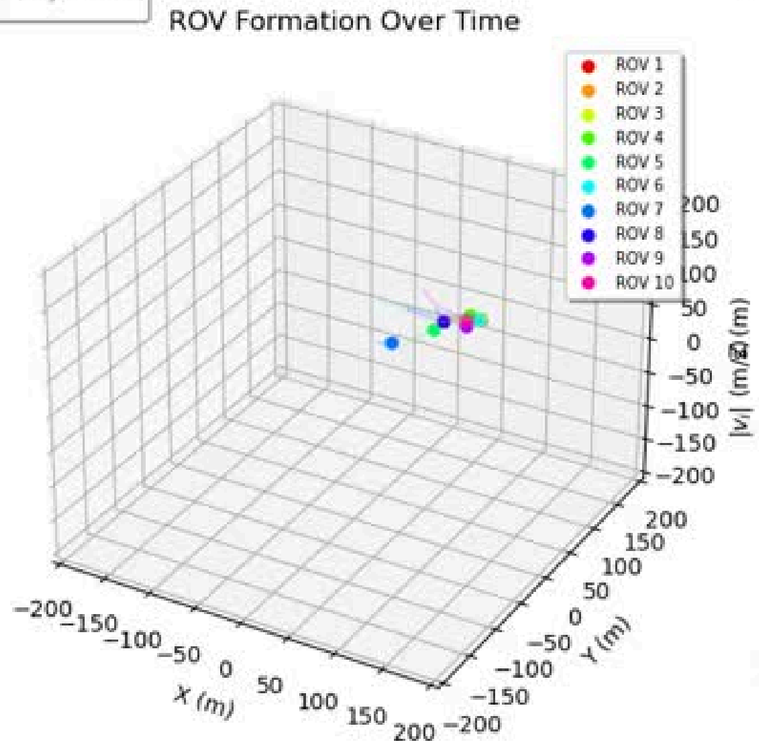
$$u_i = -c \sum_{j=1}^{n} a_{ij}(x_i - x_j)$$

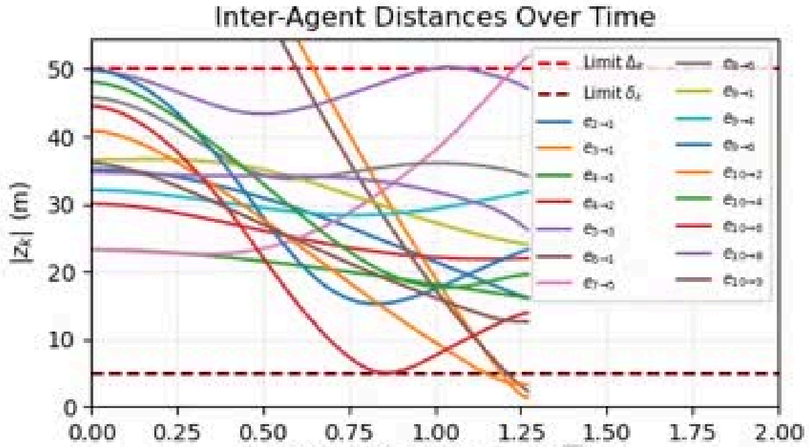
$$\dot{x} = v$$

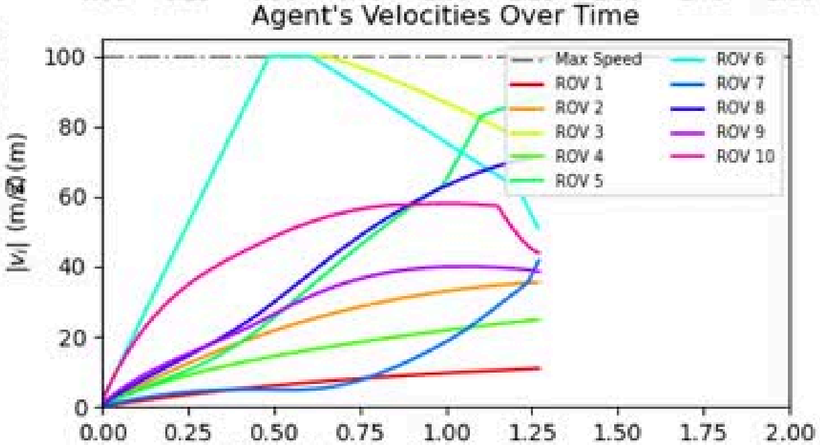
$$\dot{v} = u$$

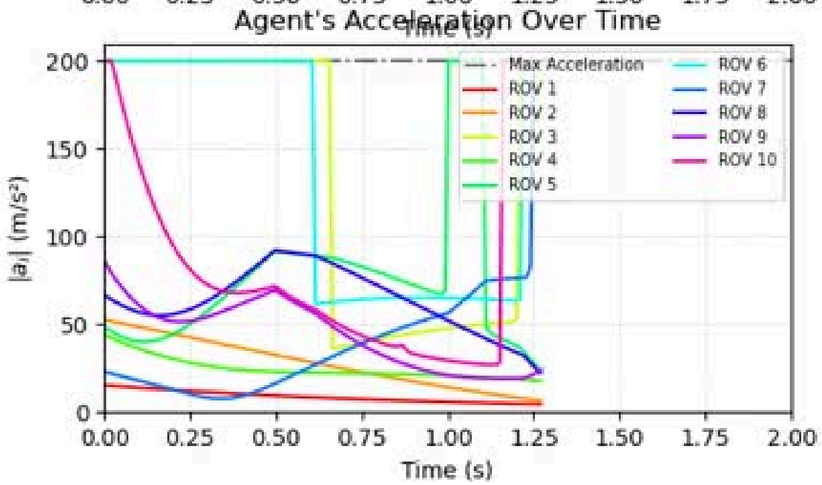
$$u_i = -c \sum_{j=1}^{n} a_{ij} \left[(x_i - x_j) + \gamma(v_i - v_j)\right]$$

```
Time:  1.27s
---------------
ROV 1 Vel: 10.90 m/s
ROV 2 Vel: 35.52 m/s
Distance to Leader: 16.06m (Target: 20.00m)
ROV 3 Vel: 65.28 m/s
Distance to Leader:  1.48m (Target: 20.00m)
ROV 4 Vel: 24.86 m/s
Distance to Leader: 16.19m (Target: 20.00m)
ROV 5 Vel: 87.74 m/s
Distance to Leader: 47.32m (Target: 20.00m)
ROV 6 Vel: 50.74 m/s
Distance to Leader:  2.38m (Target: 20.00m)
ROV 7 Vel: 41.62 m/s
Distance to Leader: 97.89m (Target: 20.00m)
ROV 8 Vel: 71.38 m/s
Distance to Leader: 36.08m (Target: 20.00m)
ROV 9 Vel: 38.58 m/s
Distance to Leader: 24.02m (Target: 20.00m)
ROV 10 Vel: 43.87 m/s
Distance to Leader: 15.69m (Target: 20.00m)
```









Next Steps

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Questions & Comments